

EVERGAME 360 SYSTEM ARCHITECTURE DIAGRAMS

Visual Reference for Technical and Executive Stakeholders

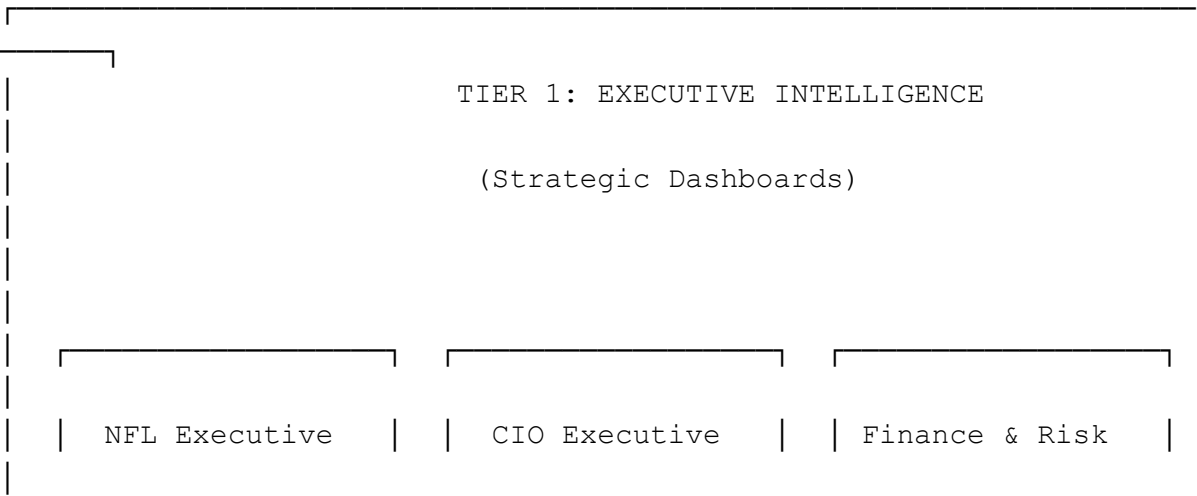
Version: 2.0

Date: November 22, 2025

Classification: Technical Documentation

1. HIGH-LEVEL SYSTEM ARCHITECTURE

Three-Tier Architecture Model



Dashboard	Dashboard	Analytics
• Readiness Map	• System Health	• Cost Tracking
• KPI Scorecard	• Vendor SLAs	• ROI Metrics
• Risk Tracking	• Cert Management	• Savings Report
• Strategic View	• Tech Monitoring	• Budget Status
[Read-Only / Strategic Access]		



Real-Time Data Feed

TIER 2: ORCHESTRATION INTELLIGENCE

(EVERGAME 360 FRAME)

Playbook Translation Engine

- Game Ops Manual → Executable Workflows

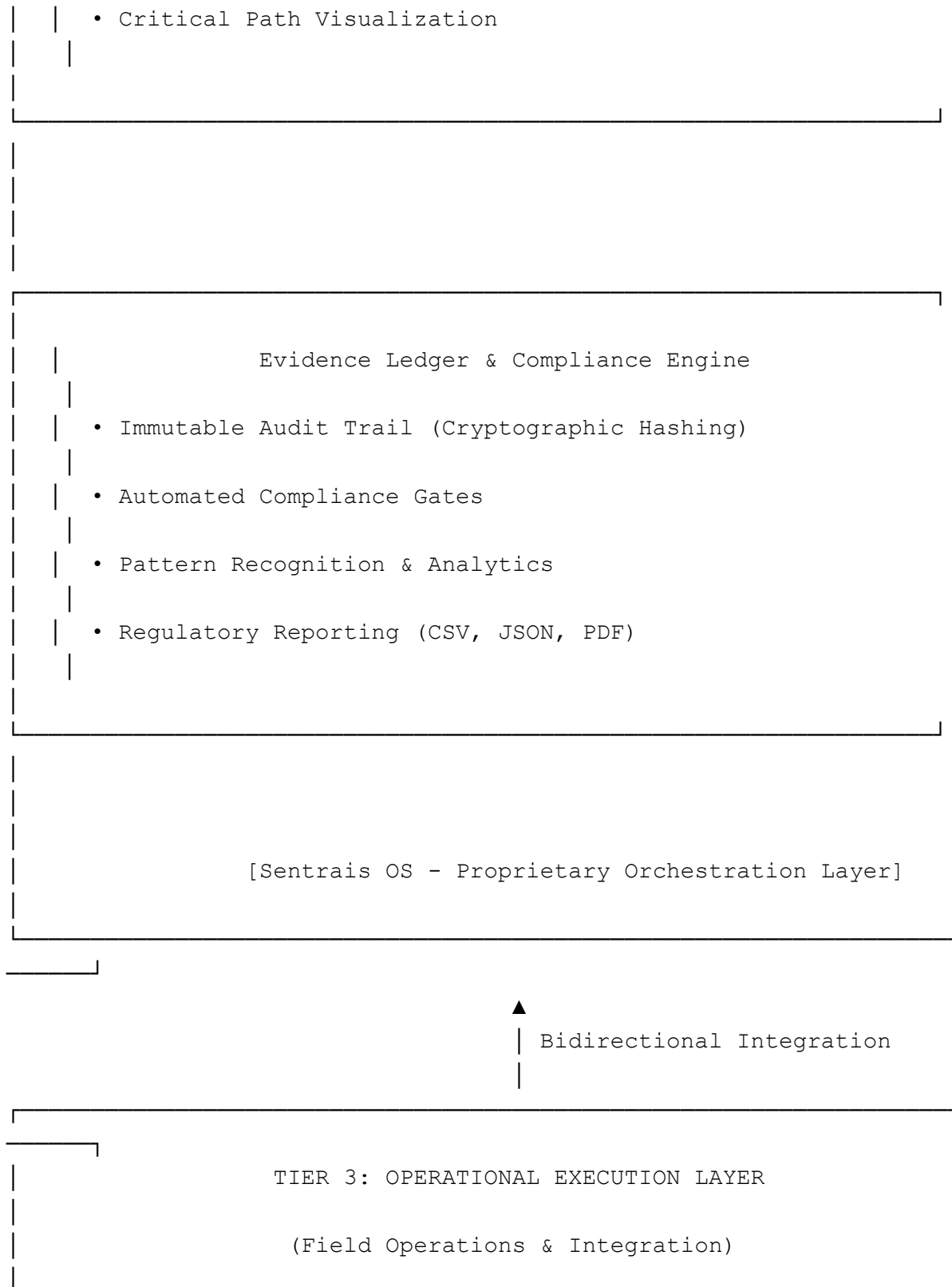
- 16 Playbooks × 721 Tasks
- State Machine Generation (BPMN 2.0)

Multi-Location Orchestration Engine

- 320+ Weekly GDA Assignments
- AI-Powered Conflict Prevention
- Real-Time Optimization (Cost, Skills, Proximity)
- Certification Validation Gates

Temporal Framework (M1-M6 Milestones)

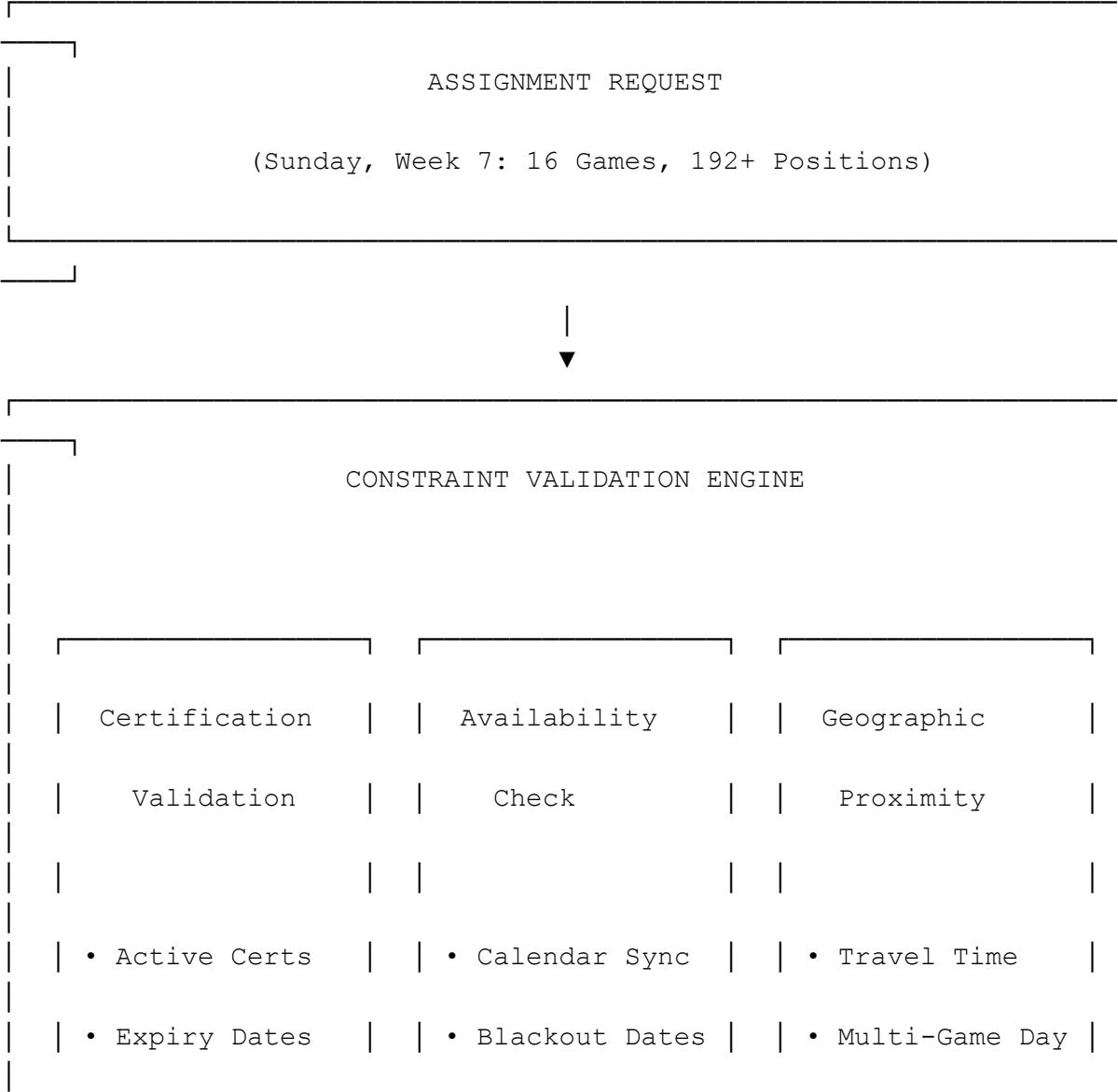
- T-48h → T-24h → T-6h → T-1h → Kickoff → +6h
- SLA Tracking & Predictive Alerts
- Dependency Chain Management



GDA Mobile	Vendor System	NFL System
Interface	Integration	Integration
• iOS App	• SVS (Video)	• UKG (Time)
• Task Lists	• C2P (Comms)	• GMS Reporting
• Evidence	• IVRS (Replay)	• iOS Ecosystem
Capture	• HAWKEYE	• WhatsApp Hub
• Offline Mode	• WiFi Systems	• Email/SMS
[Field Operators & Systems]		

2. MULTI-LOCATION ORCHESTRATION ENGINE

GDA Assignment Optimization Flow



• Multi-System	• Rest Periods	• Airport Access
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CONFLICT DETECTION & PREVENTION

Current State: 12-18 Conflicts/Week Detected Automatically

Conflict Type	Frequency	Resolution
Double-Booking	5-7/week	Auto-Alert
Certification Expired	3-4/week	Block
Travel Time Impossible	2-3/week	Suggest Alt
Rest Period Violation	1-2/week	Block
Skill Mismatch	1-2/week	Warning



AI-POWERED OPTIMIZATION

Objective Function:

Minimize: Travel Cost + Assignment Gaps + Cert Expiry Risk

Maximize: Experience Match + Load Balancing + Vendor Pref

Algorithm: Mixed-Integer Linear Programming (MILP)

Solve Time: <2 seconds for 320+ assignments

Optimality: 95%+ (vs. manual 60-70%)



ASSIGNMENT RECOMMENDATIONS

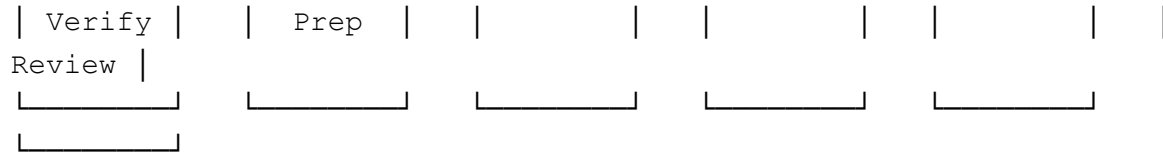
Automated Assignment (95% of cases)

- Zero Conflicts
- Optimized Cost

	• Balanced Workload	
	• → Auto-Notify GDAs	
	Manual Review Required (5% edge cases)	
	• Multiple Valid Options	
	• Special Requests	
	• VIP Games (Playoffs, Prime Time)	
	• → Ops Lead Decision with Guided Options	



	REAL-TIME MONITORING	
	• Assignment Status Dashboard	
	• Conflict Alerts (new conflicts detected post-assignment)	
	• GDA Acknowledgment Tracking	



M1: PRE-ARRIVAL VERIFICATION (T-48h to T-24h)

- └─ Certification Status Validation
- └─ Travel Arrangements Confirmed
- └─ Credential Processing
- └─ Team Assignment Briefing
- └─ Equipment Pre-Staging Check

M2: PRE-GAME PREPARATION (T-5h to T-4h)

- └─ Equipment Pickup & Inventory
- └─ Venue Access Credentials
- └─ UKG Clock-In (Geofenced)
- └─ Initial System Setup
- └─ Vendor Coordination Meeting

M3: SYSTEMS VALIDATION (T-4h to T-1h)

- └─ System Integration Tests
- └─ Network Infrastructure Validation
- └─ Vendor System Coordination
- └─ Dependency Chain Verification
- └─ Issue Triage & Escalation

M4: FINAL READINESS (T-1h to Kickoff)

- └─ 60-Minute Pre-Kickoff Check
- └─ All Systems Green Confirmation
- └─ Executive Readiness Report
- └─ Final Go/No-Go Decision
- └─ Standby Mode Activation

M5: IN-GAME OPERATIONS (Kickoff to Game End)

- └─ Live System Monitoring
- └─ Incident Response & Escalation
- └─ Real-Time Evidence Capture
- └─ Communication Hub Active
- └─ Continuous Status Updates

M6: POST-GAME REVIEW (Game End to +6h)

- └ Evidence Upload & Validation
- └ GMS.NFL.NET Report Generation
- └ Equipment Storage & Checkout
- └ UKG Clock-Out
- └ Debrief & Lessons Learned

SLA TRACKING DASHBOARD

Milestone Status: M3 (Systems Validation) - Active
Current Time: T-2h 15m

System	Tasks	Complete	In Progress	Blocked	Status
C2P	28	24	3	1	⚠
SVS	32	30	2	0	✓
IVRS	18	16	2	0	✓
EFC	12	12	0	0	✓
FTR	15	13	2	0	✓
HAWKEYE	14	12	2	0	✓
WiFi	10	9	1	0	✓
Venue Ops	22	20	2	0	✓

CRITICAL PATH ALERT:

C2P Task C2P_3.2 BLOCKED by dependency EFC_2.2 (Frequency scan incomplete)

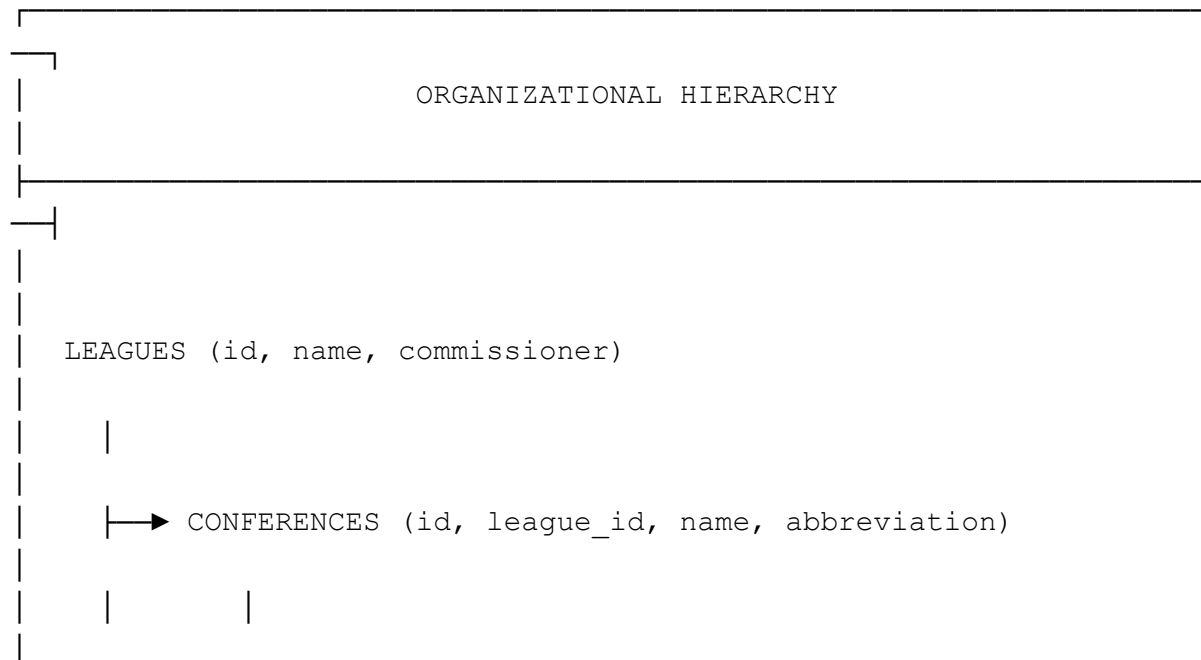
Impact: Delays final helmet module testing

Escalation: Auto-notified EFC GDA + Operations Lead

Resolution ETA: 15 minutes

4. DATA ARCHITECTURE

Database Schema (34+ Tables)



```
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|      |      |      |      |  
cert) |      |      |      |
```

DIVISIONS (id, conference_id, name)

FRANCHISES (id, division_id, name, city)

STADIUMS (id, name, capacity,

GAME OPERATIONS

```
GAMES (id, season, week, home_team, visitor_team, stadium)
```

└─> GAME_STATUS (id, game_id, milestone, readiness_score)

```
|→ GAME_INCIDENTS (id, game_id, severity, category, status)
```

↳ TASK_INSTANCES (id, game_id, task_id, gda_id, status)

└─> TASK_EVIDENCE (id, task_instance_id, file_url)

GDA MANAGEMENT

GDAS (id, name, email, phone, certifications[])

└─> GDA_CERTIFICATIONS (id, gda_id, system_id, expiry)

└─> GDA_ASSIGNMENTS (id, game_id, gda_id, role, status)

└─> GDA_AVAILABILITY (id, gda_id, date, available, reason)

SYSTEM & TASK ORCHESTRATION

```
SYSTEMS (id, name, vendor, playbook_count, total_tasks)
```

```
|
```

```
└─▶ PLAYBOOKS (id, system_id, name, role, location)
```

```
|
```

```
|
```

```
| └─▶ TASKS (id, playbook_id, sequence, description,
```

```
|
```

```
        nin_phase, milestone, dependencies[])
```

COMPLIANCE & AUDIT

```
COMPLIANCE_RULES (id, rule_type, severity, enforcement_logic)
```

```
|
```

```
└─▶ COMPLIANCE_VIOLATIONS (id, game_id, rule_id, status)
```

```
EVIDENCE_LEDGER (id, event_type, data_hash, timestamp, signature)
```


AUDIT_TRAIL (id, user_id, action, entity_type, changes)

ANALYTICS & REPORTING

READINESS_SCORES (id, game_id, milestone, score, components)

PERFORMANCE_METRICS (id, franchise_id, season, kpi, value)

VENDOR_PERFORMANCE (id, vendor_id, system_id, sla_compliance)

FINANCIAL_ANALYTICS (id, game_id, labor_cost, roi_impact)

DATA FLOW: Write-Once, Read-Many (Optimized for Analytics)

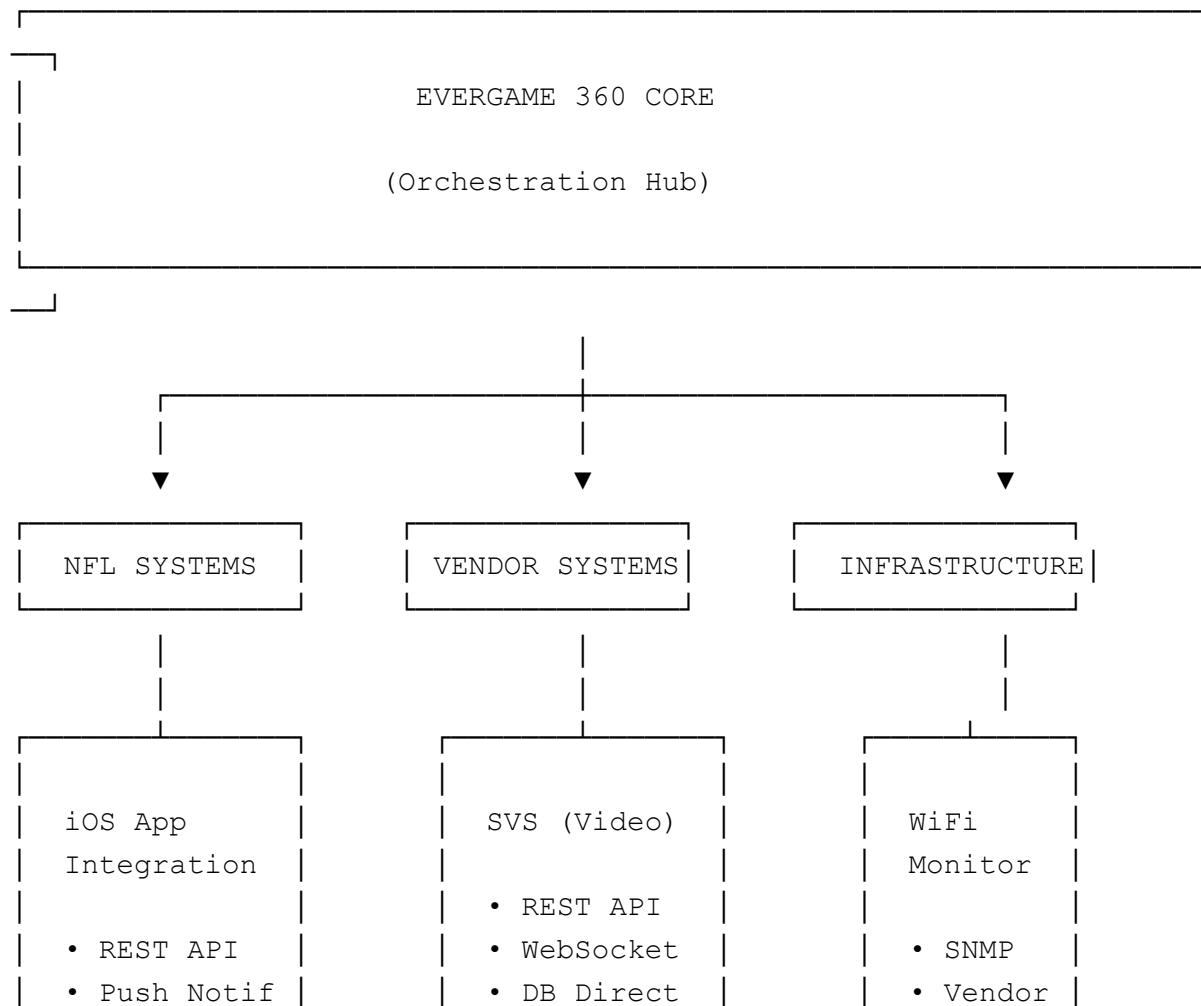
— Operational Data: Real-time updates via WebSocket

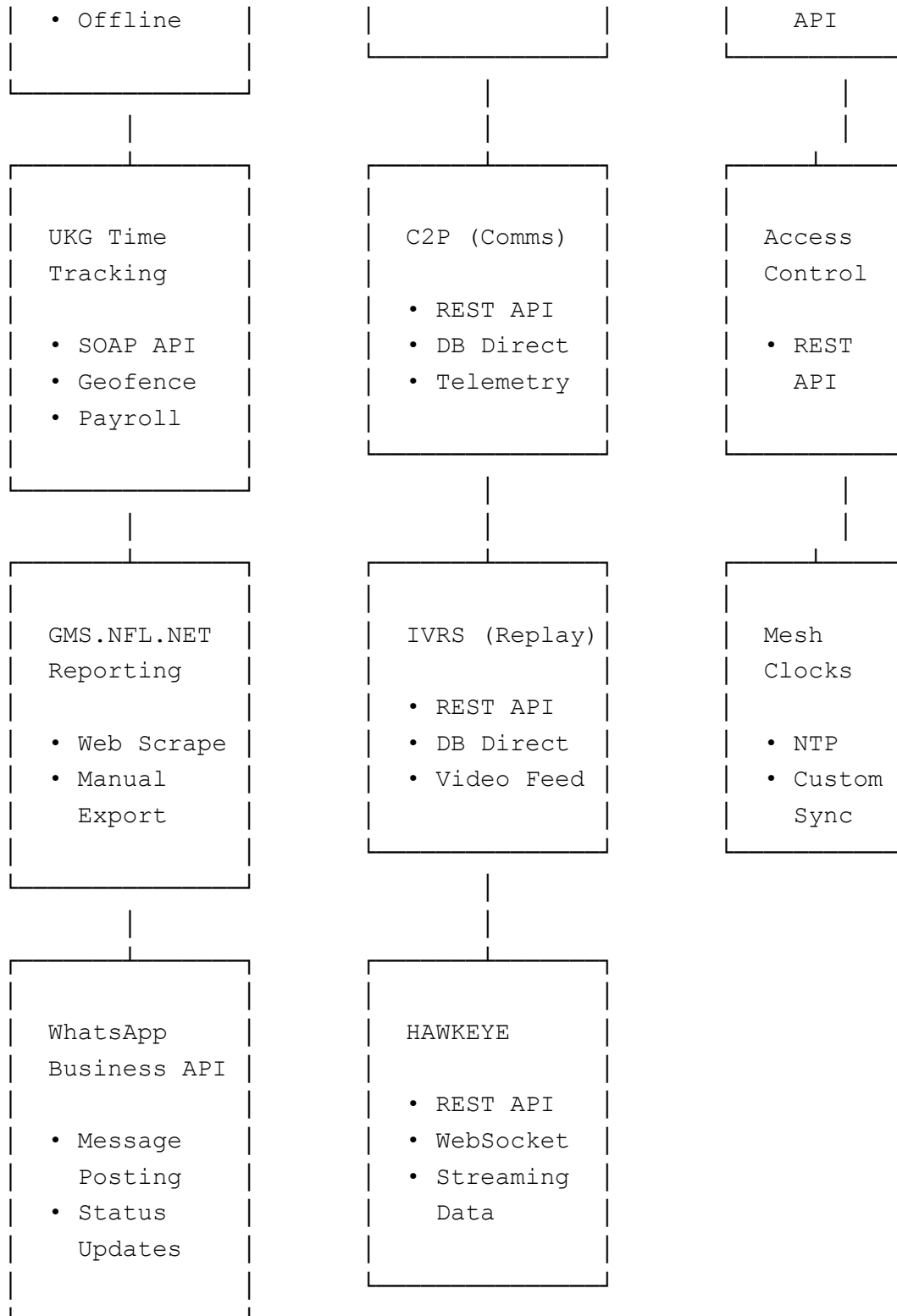
— Evidence Ledger: Append-only with cryptographic hashing

- └ Analytics: Pre-computed aggregations, cached results
- └ Reporting: Materialized views, indexed for executive queries

5. INTEGRATION ARCHITECTURE

System Integration Map



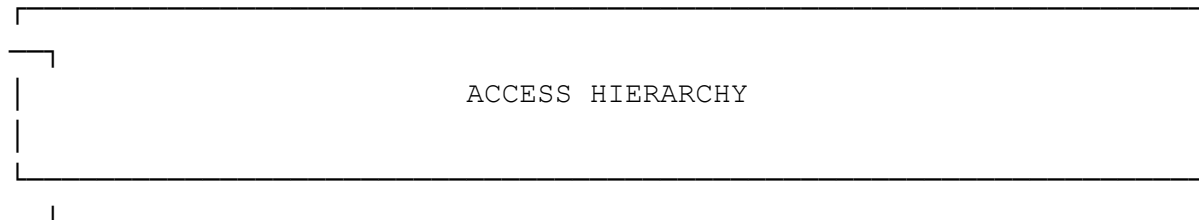


INTEGRATION PATTERNS:

-
1. API Integration (Preferred)
 - Modern REST/GraphQL APIs
 - 100-500ms latency
 - Real-time updates
 2. Database Integration
 - Legacy systems without APIs
 - Read-only queries
 - 1-5s latency
 3. File/Message Queue
 - Batch processing
 - RabbitMQ/Kafka
 - Non-real-time data
 4. WebSocket Streaming
 - Bidirectional real-time
 - Live operations data
 - <100ms latency

6. SECURITY & ACCESS CONTROL

Role-Based Access Control (RBAC)



LEVEL 1: NFL EXECUTIVE LEADERSHIP

Commissioner, COO, CFO
Access: Read-Only League-Wide
Scope: All Franchises, Aggregated Data
✓ Strategic Dashboards
✓ KPI Scorecards
✓ Compliance Reports
✓ Financial Analytics
✗ Operational Edits
✗ Individual GDA Details



LEVEL 2: CIO / IT EXECUTIVE

Chief Information Officer
Access: Monitoring + Limited Actions
Scope: Technical Systems, Vendor Data
✓ System Health Dashboards
✓ Vendor Performance Metrics
✓ Integration Monitoring
✓ Emergency Overrides
✓ Certification Management
✗ Playbook Content Editing



LEVEL 3: NFL OPERATIONS LEAD

Football Technology, Game Operations VP
Access: Full Edit Authority
Scope: All Playbooks, Assignments, Rules
✓ Playbook Management (CRUD)
✓ GDA Assignment Control

- ✓ Compliance Rule Configuration
- ✓ Operational Command Center
- ✓ Custom Reporting & Analytics
- ✓ Exception Approval Workflows



LEVEL 4: FRANCHISE LEADERSHIP

- Club President, CTO, Game Day Designee
Access: Read-Only Franchise-Specific
Scope: Own Franchise Games & Performance
- ✓ Franchise Game Dashboards
 - ✓ GDA Performance (franchise games)
 - ✓ Vendor SLA Compliance
 - ✓ Incident Reports (own games)
 - ✗ Other Franchise Data
 - ✗ League-Wide Aggregations



LEVEL 5: GDA (FIELD OPERATORS)

- Orange Hat, Purple Hat, Blue Hat, etc.
Access: Execution Only (Time-Gated)
Scope: Assigned Tasks During T-6h to +6h
- ✓ Task List (assigned games only)
 - ✓ Evidence Capture (photo, notes)
 - ✓ System Integration (vendor systems)
 - ✓ Communication (WhatsApp, team chat)
 - ✗ Other GDA Assignments
 - ✗ Historical Data
 - ✗ Outside Operational Window



LEVEL 6: VENDOR PARTNERS

SVS, C2P, IVRS, HAWKEYE
Access: System-Specific Data
Scope: Own System Performance Metrics
✓ Integration Endpoints (API access)
✓ Performance Metrics (own system)
✓ Incident Reports (own system)
✓ SLA Compliance Dashboard
✗ Other Vendor Data
✗ GDA Personal Information

DATA ISOLATION: Franchise VPC Boundaries

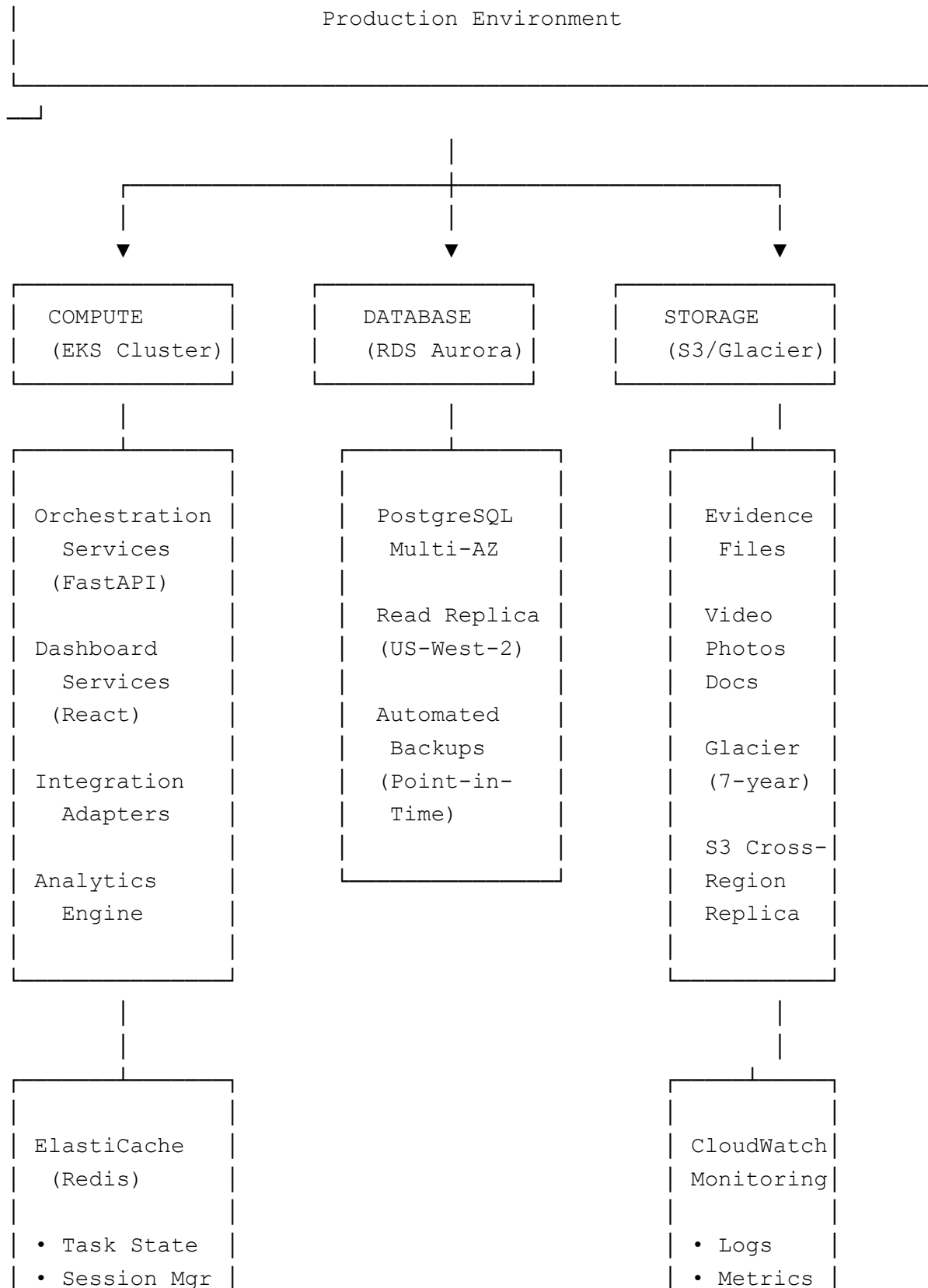
Each franchise data resides in dedicated Virtual Private Cloud (VPC):

- Network-level isolation
- Separate encryption keys
- Independent backup schedules
- Cross-franchise queries only via aggregation service

7. DEPLOYMENT ARCHITECTURE

Cloud Infrastructure (AWS Multi-Region)

US-EAST-1 (PRIMARY)



- Real-Time Updates

- Alerts
- Dashbd

US-WEST-2 (DR/FAILOVER)

Disaster Recovery Environment

- Standby EKS Cluster (Hot Standby)
- RDS Read Replica (Promoted to Primary on Failover)
- S3 Cross-Region Replication (Real-Time)
- Automatic Failover: RTO 15min, RPO 5min

SECURITY LAYER

- WAF (Web Application Firewall) - DDoS Protection
- VPC Isolation (Per Franchise + Shared Services)
- IAM (Role-Based Access Control)
- KMS (Encryption at Rest - AES-256)
- TLS 1.3 (Encryption in Transit)
- Security Groups (Network ACLs)
- GuardDuty (Threat Detection)
- CloudTrail (Audit Logging)

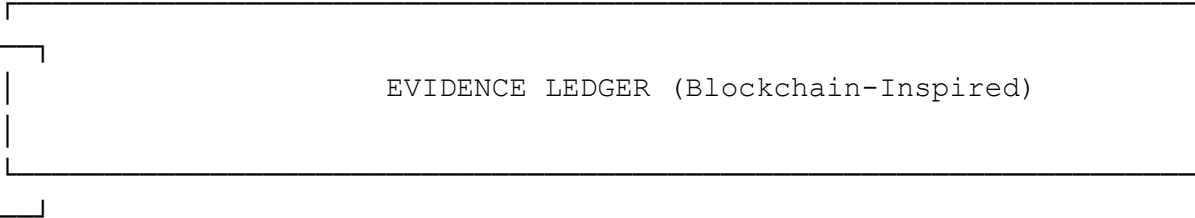
SCALABILITY & PERFORMANCE

- Horizontal Auto-Scaling: 10-100+ pods based on load
- Load Balancing: Application Load Balancer (ALB)

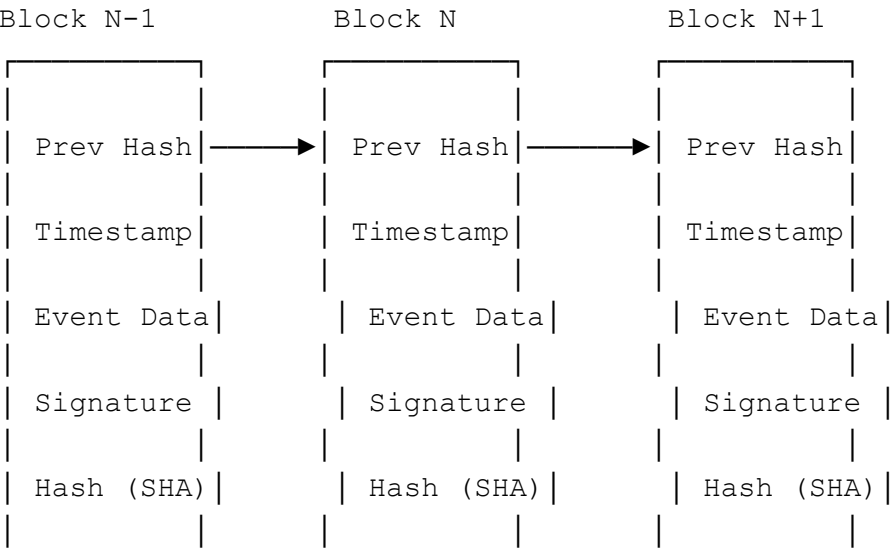
- CDN: CloudFront for static assets (dashboard UI)
- Caching: Multi-layer (CDN, Redis, Application)
- Database Sharding: Future support for 100+ franchises

8. EVIDENCE LEDGER ARCHITECTURE

Immutable Audit Trail



APPEND-ONLY DATA STRUCTURE



CAPTURED EVIDENCE TYPES

Event Type	Captured Data
Task Completion	Who, What, When, Where, Evidence (photo/video)
GDA Assignment	Assignment change, Authorized by, Reason
System Config Change	Playbook edit, Change details, Approver
Compliance Gate Validation	Rule validated, Pass/Fail, Evidence required
Incident Report	Severity, Category, Resolution, Impact
Communication Log (Metadata)	Metadata only (not content) , Timestamp, Parties

COMPLIANCE EXPORT FORMATS

Format	Use Case
CSV	Excel analysis, Spreadsheet import
JSON	API integration, System-to-system
PDF Report	Executive summaries, Board presentations

CJIS/SOC2	Regulatory audit
Format	submissions

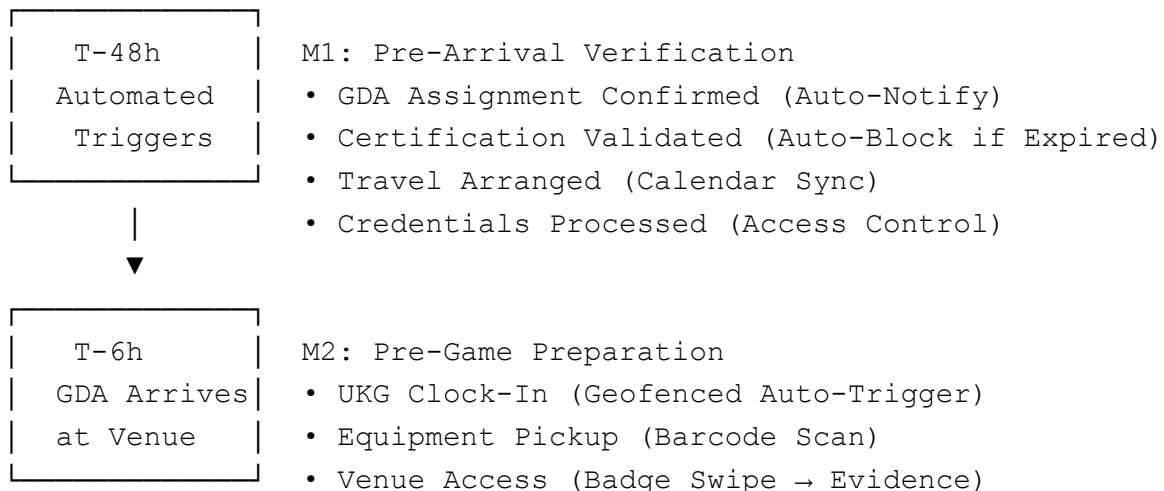
QUERY INTERFACE

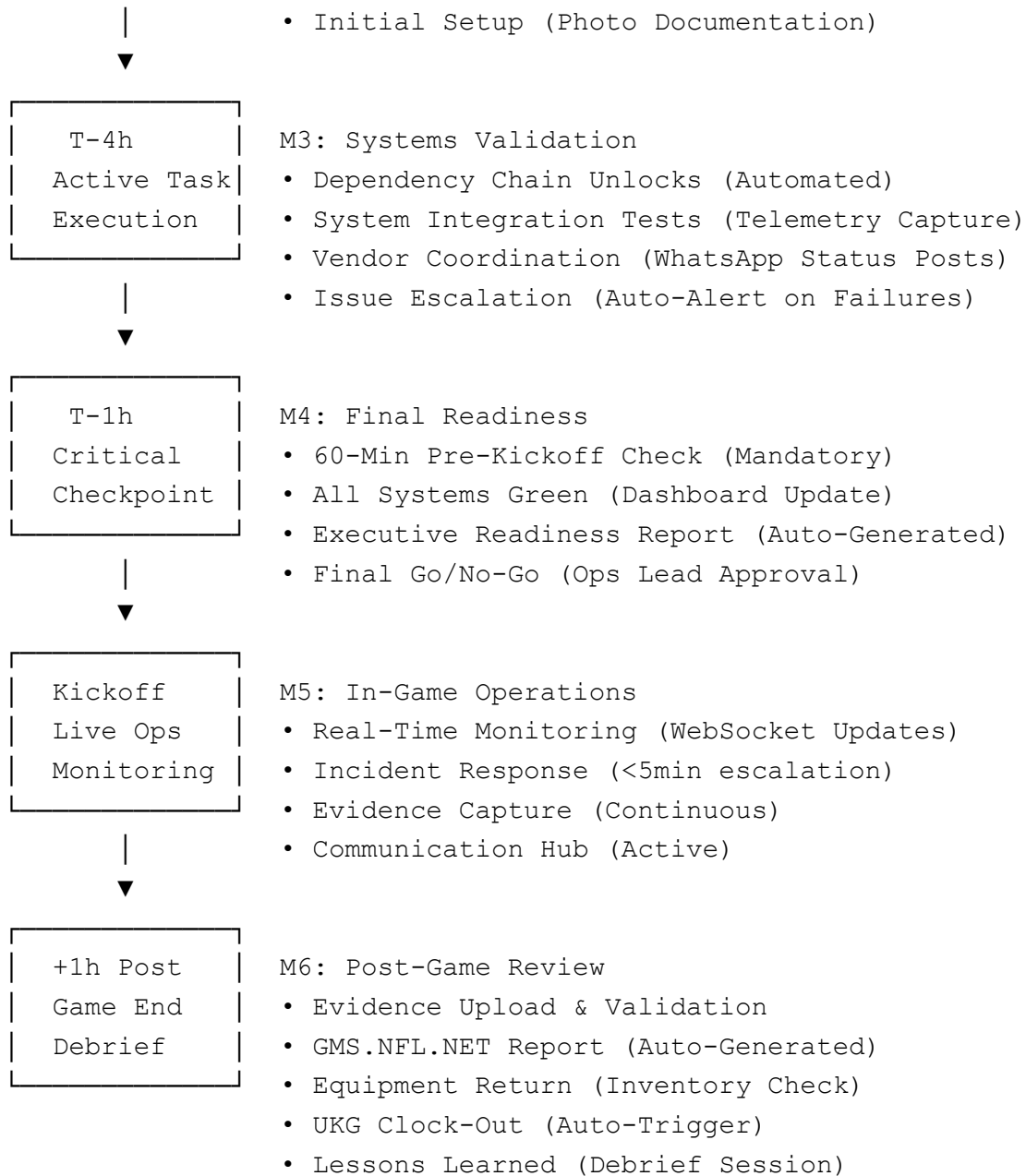
- After-Action Review: "Show all incidents from Week 7"
- Compliance Audit: "Export all C2P compliance validations Q4"
- Performance Analysis: "Compare GDA completion rates by franchise"
- Legal Discovery: "All evidence for Game ID 2024-REG-W12-DAL-PHI"

9. OPERATIONAL WORKFLOW

End-to-End Game Day Flow

PRE-GAME WORKFLOW (T-48h to Kickoff)





REAL-TIME DASHBOARD UPDATES

Every task completion → Evidence Ledger entry → Dashboard refresh (<1s)

EXECUTIVE VISIBILITY

CIO Dashboard: Live M1-M6 progress across all games

NFL Executive: Weekly readiness heatmap (32 franchises × week's games)

CONCLUSION

This architecture delivers:

- ✓ **Executive Intelligence:** Real-time visibility into league-wide operations
- ✓ **Operational Excellence:** 99.5% readiness through automated orchestration
- ✓ **Financial Impact:** \$3.25M annual net benefit, 340% ROI
- ✓ **Risk Mitigation:** 70% compliance improvement, incident prevention
- ✓ **Scalability:** Supports league expansion, international games, playoffs

EVERGAME 360 transforms NFL game day operations from reactive crisis management to proactive intelligence orchestration.

Prepared by NOVATELabs

Visual Architecture Reference - Version 2.0

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